

Introduction to Probability and Applied Statistics

Week 4: Expectation

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Roadmap

Discrete distributions

Continuous distributions

Data-generating processes

Conditioning

Expectation

Variance and covariance

Wrapping up

The probability mass function

PMF

The **probability mass function** of a discrete random variable X is

$$f(x) = \mathbb{P}(X = x).$$

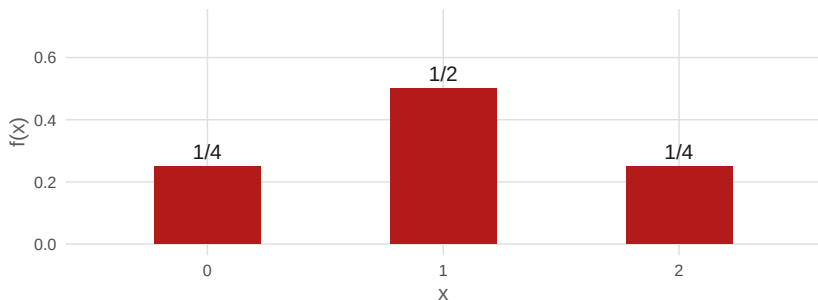
The **support** of X is the set of values at which $f(x) > 0$.

$X = x$ names an **event**: the set of outcomes to which X assigns the number x , written $\{s \in \Omega : X(s) = x\}$. Taking its probability is therefore defined.

$\mathbb{P}(X)$ is undefined. A probability attaches to an event, never to a random variable.

Two coin tosses

X is the number of heads in two tosses of a fair coin,
 $\Omega = \{HH, HT, TH, TT\}$.



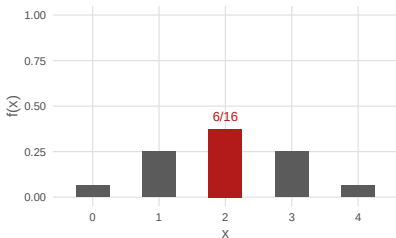
$f(0) = \frac{1}{4}$, $f(1) = \frac{1}{2}$, $f(2) = \frac{1}{4}$, and $f(x) = 0$ at every other x .

Let $Y = 2 - X$, the number of tails. Y has the same PMF as X . A PMF does not determine the random variable: X and Y are different functions on the same sample space.

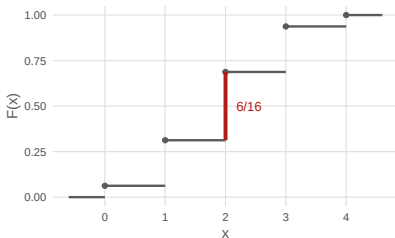
From PMF to CDF, and back

Now X is the number of heads in **four** tosses. There are $2^4 = 16$ equally likely sequences.

PMF $f(x)$



CDF $F(x)$



$F(x)$ sums the PMF over every value at or below x :

$$F(1) = f(0) + f(1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}.$$

The height of the jump in F at x is $f(x)$: $f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{6}{16}$.

What a PMF must satisfy

PROPERTIES

1. $0 \leq f(x) \leq 1$ for every x ;

Each $f(x)$ is the probability of the event $X = x$. No probability is negative; none exceeds 1.

2. $\sum_x f(x) = 1$;

On every draw X takes exactly one value from the support. Summing over the support therefore totals 1.

3. $\mathbb{P}(X \in A) = \sum_{a \in A} f(a)$ for any set of values A .

One outcome produces one value of X , so different values are disjoint events. Property 3 is the addition rule applied to them.

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From discrete to continuous

DISCRETE

The support is countable.

Probability is a property of points.

Examples: seats won, battle deaths, vetoes cast, protests in a city-month.

Probability mass function (PMF)

$$f(x) = \mathbb{P}(X = x)$$

CONTINUOUS

The support is an interval.

$\mathbb{P}(X = x) = 0$ for every x , so probability is a property of intervals.

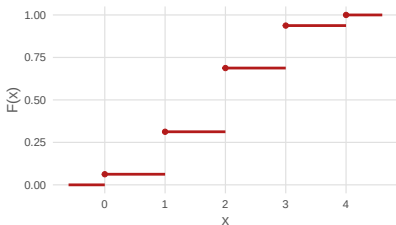
Examples: vote share, GDP per capita, time until a government falls.

Probability density function (PDF)

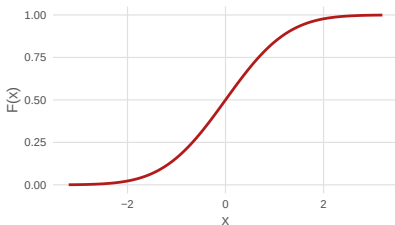
$$f(x) = F'(x)$$

Continuous random variables: Why $\mathbb{P}(X = x) = 0$

Discrete



Continuous



The height of a jump in F at x is $\mathbb{P}(X = x)$.

The CDF of a continuous random variable has no jumps. So $\mathbb{P}(X = x) = 0$ at every x .

A PMF for such a variable would equal 0 everywhere. A density is used in its place.

Continuous random variables

For a continuous random variable the probability of any single value is zero. Probability is a property of intervals instead:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

The **probability density function** $f(x)$ is the derivative of the CDF $F(x)$, and the limit of a histogram as the bins narrow.

A valid PDF has $f(x) \geq 0$ everywhere and encloses a total area of 1:
 $\int_{-\infty}^{\infty} f(x) dx = 1$.

$f(x)$ is not a probability, and $f(x) > 1$ is possible. A probability is obtained by integrating f over an interval.

The endpoints carry probability 0, so they may be included or excluded without changing the answer: $\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X \leq b)$.

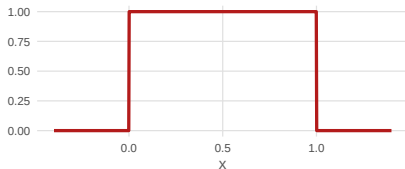
Continuous Uniform Distribution

UNIFORM

$X \sim \mathcal{U}(a, b)$ is a completely random number between a and b : the density is *constant* over the interval.

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

PDF



The area under the density is a rectangle, width $b - a$, height $1/(b - a)$. **Probability is proportional to length**: a subinterval of length ℓ has probability $\ell/(b - a)$.

The **standard uniform** $\mathcal{U}(0, 1)$ is the one we use most: $f(x) = 1$ on the unit interval.

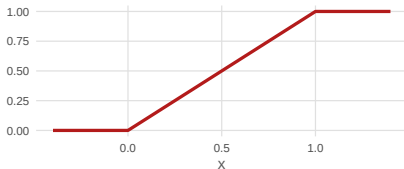
The uniform CDF

UNIFORM CDF

Integrating the constant density gives the CDF of $X \sim \mathcal{U}(a, b)$:

$$F(x) = \begin{cases} 0 & \text{if } x \leq a, \\ \frac{x-a}{b-a} & \text{if } a < x < b, \\ 1 & \text{if } x \geq b. \end{cases}$$

CDF



$F(x)$ is the area under the density to the left of x . The density is constant, so that area grows at a constant rate: a straight line from 0 at a to 1 at b , slope $1/(b - a)$.

For the standard uniform, $F(x) = x$. A draw falls below 0.3 with probability 0.3.

Standard Normal Distribution

STANDARD NORMAL

A continuous random variable Z has the **standard normal distribution** if its PDF is

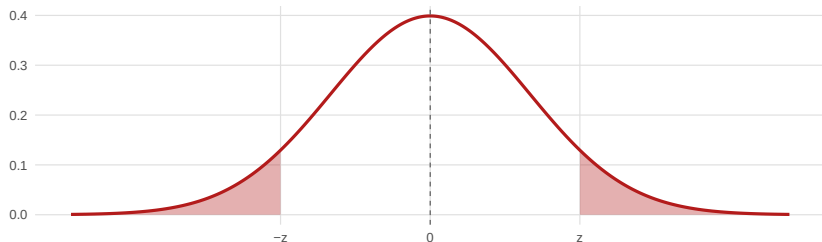
$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

We write $Z \sim \mathcal{N}(0, 1)$.

The $1/\sqrt{2\pi}$ in front is a **normalizing constant**. It is there for one reason: to make the total area under the PDF equal 1. It sets the height of the curve, not its shape.

The standard normal is used so often that its PDF and CDF have their own Greek letters: φ for the PDF, Φ for the CDF. Z is the conventional name for the variable itself.

Symmetry of the standard normal



- **Symmetry of the PDF:** $\varphi(z) = \varphi(-z)$, so the curve is a mirror image about 0 .
- **Symmetry of tail areas:** the area left of $-z$ equals the area right of z , which gives $\Phi(z) = 1 - \Phi(-z)$.
- **Symmetry of Z and $-Z$:** if $Z \sim \mathcal{N}(0, 1)$, then $-Z \sim \mathcal{N}(0, 1)$ as well.

From the standard normal to the Normal

NORMAL

If $Z \sim \mathcal{N}(0, 1)$, then

$$X = \mu + \sigma Z$$

has the **Normal distribution** with mean μ and variance σ^2 , for any real μ and σ^2 with $\sigma > 0$. We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

Shifting by μ moves the center. Scaling by σ stretches the spread. Every normal is the standard normal moved and stretched.

Recall that s is the sample standard deviation, calculated from the data. Here σ is the population standard deviation. Note that we multiply Z by σ , not by the variance σ^2 ; else the units would be wrong and X would have variance σ^4 .

Standardization

Of course, if we can get from $\mathbf{Z}$ to $\mathbf{X}$, then we can get from $\mathbf{X}$ back to $\mathbf{Z}$. The process of getting a standard Normal from a non-standard Normal is called, appropriately enough, *standardization*.

For $\mathbf{X} \sim \mathcal{N}(\mu, \sigma^2)$, the *standardized version* of $\mathbf{X}$ is

$$\frac{\mathbf{X} - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Normal Distribution

NORMAL

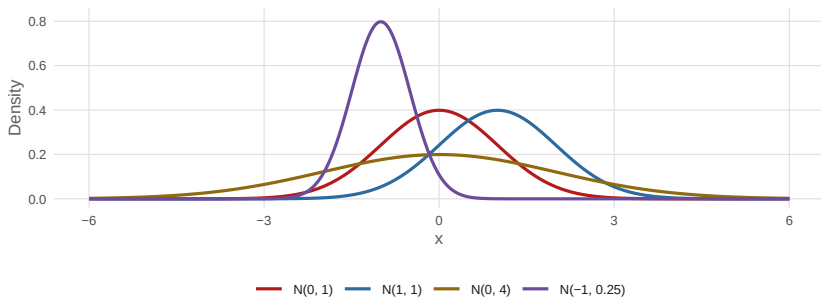
$X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right).$$

The case $\mu = 0, \sigma^2 = 1$ is the **standard normal**.

Recall that s^2 is the sample variance, calculated from the data. Here σ^2 is the population variance.

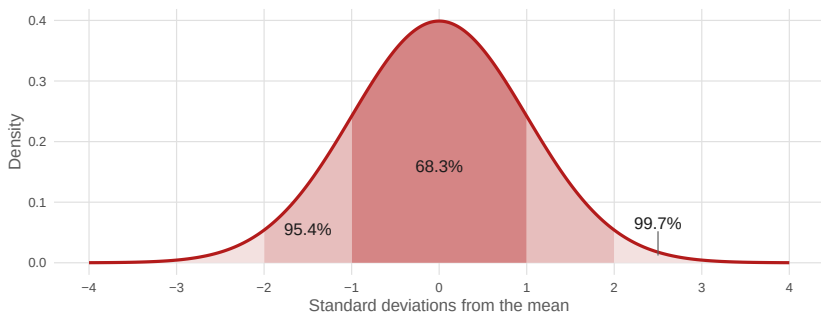
Normal Distribution



$\mathcal{N}(-1, 0.25)$ moves both: left of 0 and narrower.

Symmetric about μ , so the mean, the median and the mode coincide.

The standard normal, $\mathcal{N}(0, 1)$



You'll see this again, as the 68–95–99.7 rule. In a normally distributed population, ~68% of observations fall within one standard deviation of the mean, 95% within two, 99.7% within three.

Reading the normal in R

```
pnorm(1) - pnorm(-1)          # P(-1 <= X <= 1) for a standard normal
```

```
[1] 0.6826895
```

```
qnorm(0.975)                  # the value with 97.5% of the mass below it
```

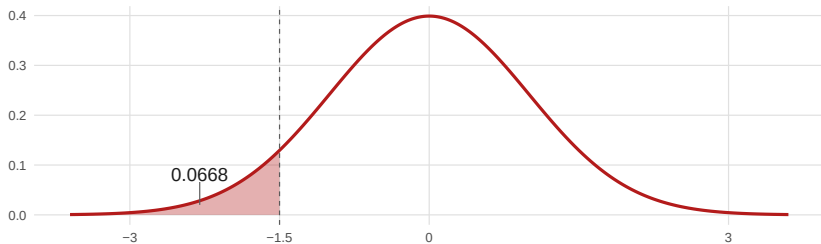
```
[1] 1.959964
```

`p` gives the CDF, `q` its inverse, `d` the density, `r` a random draw. Every distribution in R follows the same four-letter convention: `pnorm`, `dbinom`, `rpois`, `qunif`.

pnorm: the area to the left

```
pnorm(-1.5)
```

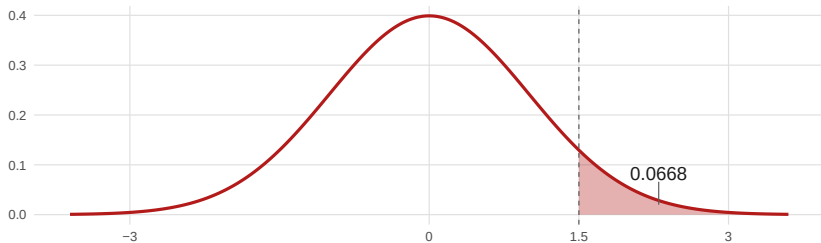
```
[1] 0.0668072
```



pnorm: the area to the right

```
pnorm(1.5, lower.tail = FALSE)
```

```
[1] 0.0668072
```

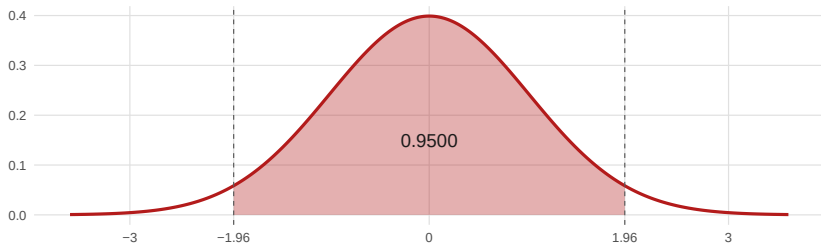


1 - pnorm(1.5) gives the same number.

pnorm: the area between

```
pnorm(1.96) - pnorm(-1.96)
```

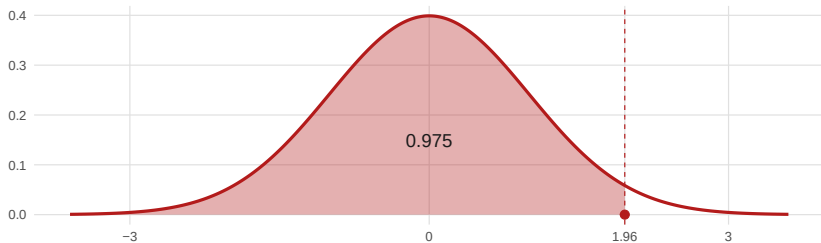
```
[1] 0.9500042
```



qnorm: from probability back to cutoff

```
qnorm(0.975)
```

```
[1] 1.959964
```



`pnorm` takes the cutoff and returns the area. `qnorm` takes the area and returns the cutoff.

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The data-generating process

DATA-GENERATING PROCESS

The **data-generating process** is the mechanism, in the world and in the study design, that produced the rows you have.

Gailmard calls it the set of rules that produce data.

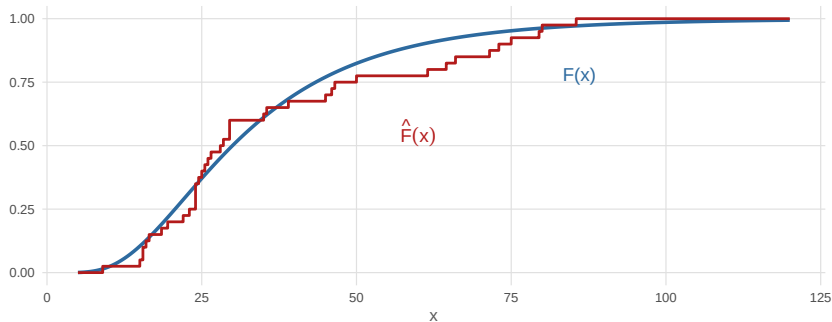
You never observe it. You observe one dataset it produced.

What is the DGP?

Your data looks like this:

pink flower	green leaf	ants
orange flower	yellow leaf	chopstick paper
hot pink flower	stick	soy sauce packet
green flower	stick	wrapper

The process fixes F ; the sample gives you $\hat{F}$



$\hat{F}(x)$ is the empirical CDF (eCDF).

Components of a process

Every part of a process is either fixed or the product of chance.

- **Deterministic**: typically 'fixed' or 'given'
- **Probabilistic**: product of a probabilistic process

These probabilistic components can themselves be of different types:

- **Systematic**: product of other variables, of which at least one is probabilistic
- **Stochastic**: the randomness that remains once the systematic part is fixed

A process for turnout

```
n    <- 5000                                # deterministic: the frame
age  <- sample(18:85, n, replace = TRUE)    # stochastic
ed   <- rbinom(n, 1, 0.32)                  # stochastic

p    <- 0.20 + 0.004 * (age - 18) + 0.15 * ed # systematic
V    <- rbinom(n, 1, p)                      # stochastic

sum(V)                                     # how many voted
```

```
[1] 1910
```

`sample()` draws from a discrete uniform, `rbinom(n, 1, .)` from a Bernoulli.

What each line is

Object	Component	Why
n , the 18+ rule	Deterministic	Fixed by the design, not drawn
age, ed	Stochastic	Drawn, independent of the rest
p	Systematic	A function of age and ed
v	Stochastic	A Bernoulli draw, given p

The survey's part of the process

```
p_lie    <- 0.08 + 0.12 * ed                # systematic  
reported <- ifelse(V == 1, 1, rbinom(n, 1, p_lie)) # stochastic
```

A non-voter without a degree reports voting with probability .08. A non-voter with a degree reports voting with probability .20.

`p_lie` is systematic: a function of `ed`. `reported`, not `V`, is what you are handed.

Ansolabehere and Hersh (2012), "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate," *Political Analysis* 20(4).

The DGP includes...

WHAT HAPPENS IN THE WORLD

People really do differ in age and education, and those differences really do bear on whether they vote.

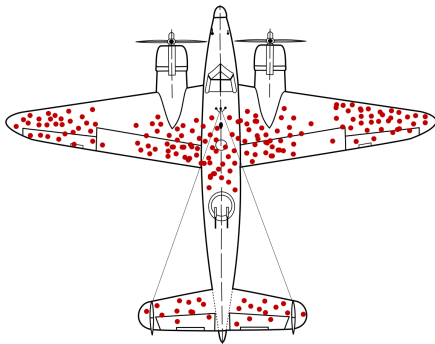
The object of inquiry.

WHAT HAPPENS IN THE STUDY

The sampling frame, the recruitment, the question wording, who picked up the phone, who told the truth.

Both are in the data-generating process. Only one of them is what you set out to learn about, and no amount of sample size separates them.

Where the holes are



The returning aircraft are a sample, and returning is part of the process that produced it. Something determined which rows you got to see.

McGeddon / Wikimedia Commons, CC BY-SA 4.0

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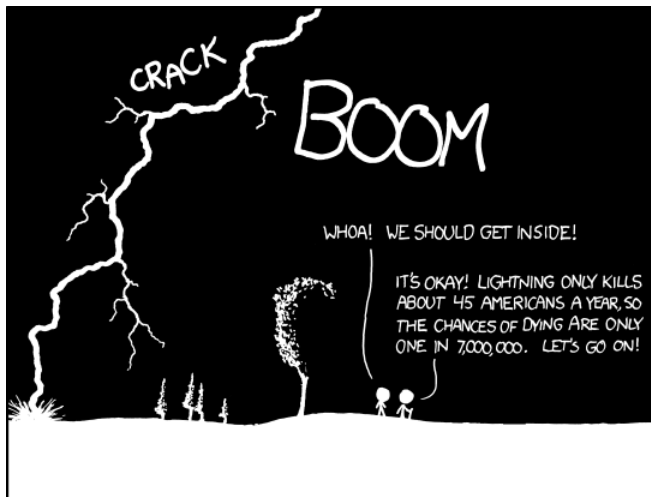
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Conditional risk

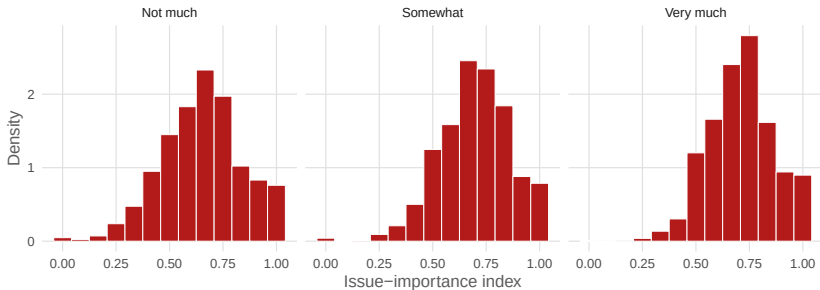


THE ANNUAL DEATH RATE AMONG PEOPLE
WHO KNOW THAT STATISTIC IS ONE IN SIX.

Conditional distributions

Y is the issue-importance index (horizontal axis). X is political interest.

Political interest:



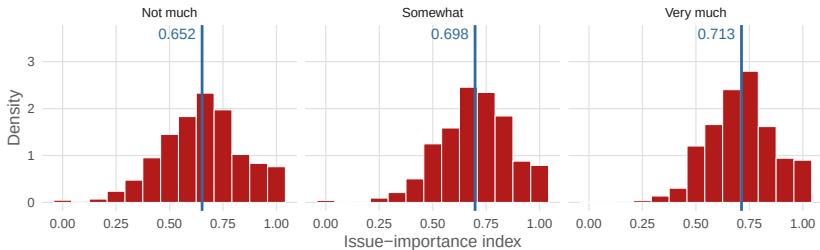
One distribution per value of X , each normalized to area 1.

The conditional mean

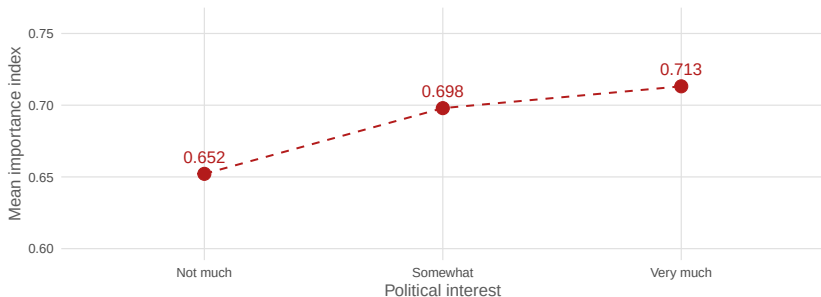
Y is the issue-importance index (horizontal axis). X is political interest.

$$\bar{Y}_{X=x} = \frac{\sum_i Y_i \mathbf{1}\{X_i = x\}}{\sum_i \mathbf{1}\{X_i = x\}}$$

Political interest:



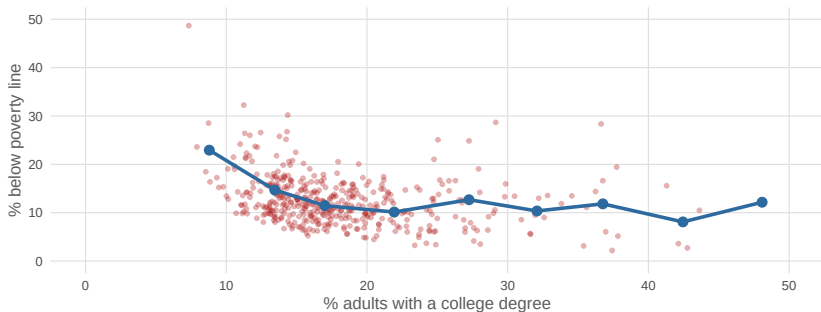
The conditional mean function



$\bar{Y}(x)$: the conditional mean, read as a function of x .

Conditioning on a continuous variable

Recall the 437 Midwestern counties from Week 2.



With 437 distinct x values, group into bins and take the mean of y in each.

The overall mean is a weighted average of group means

For the same 437 counties, let's consider poverty by metro status (`inmetro` = 1 if the county lies in a Census metropolitan area).

```
groups <- midwest |>
  group_by(inmetro) |>
  summarize(ybar = mean(percbelowpoverty), n = n()) |>
  mutate(phat = n / sum(n))
groups
```

```
# A tibble: 2 x 4
```

	inmetro	ybar	n	phat
	<int>	<dbl>	<int>	<dbl>
1	0	13.67	287	0.6568
2	1	10.29	150	0.3432

The overall mean is a weighted average of group means

```
groups |>
  summarize(weighted_by_group_size = sum(ybar * phat),
            overall_mean           = mean(midwest$percbelowpoverty),
            unweighted_of_the_2    = mean(ybar))
```

```
# A tibble: 1 x 3
  weighted_by_group_size overall_mean unweighted_of_the_2
      <dbl>           <dbl>           <dbl>
1      12.51         12.51         11.98
```

$$\bar{Y} = \sum_x \bar{Y}_{X=x} \hat{p}(x)$$

“The **empirical** law of iterated expectations”

Note: 12.51 is the mean of county poverty rates. It is not the share of Midwesterners below the poverty line. For that, each county would need to be weighted by its population (`poptotal`). What the weights should be depends on which unit you want to count equally: counties, groups or people.

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Expected value

EXPECTATION

The **expected value** of X , written $\mathbb{E}[X]$, is

$$\mathbb{E}[X] = \sum_{i=1}^k p_i x_i$$

for a discrete X taking the values $x_1, \dots, x_k$ with $p_i = \mathbb{P}(X = x_i)$.

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

for a continuous X with density f .

Expectation is a weighted average

$$\bar{X} = \underbrace{\sum_j x_j \hat{p}(x_j)}_{\text{Week 2}}$$

$$\mathbb{E}[X] = \underbrace{\sum_j x_j \mathbb{P}(X = x_j)}_{\text{this week}}$$

The same sum. The observed relative frequency $\hat{p}$ is replaced by the probability $\mathbb{P}$.

Expectation of a fair die

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

An expected value is a property of a distribution. Membership in the support is a separate question (3.5 is not a value X can take).

Expectation of a Bernoulli variable

$X \sim \text{Bernoulli}(p)$ takes the value 1 with probability p and 0 with probability $1 - p$:

$$\mathbb{E}[X] = 0(1 - p) + 1(p) = p$$

At $p = .6$, $\mathbb{E}[X] = .6$.

Expectation of a standard normal

$Z \sim \mathcal{N}(0, 1)$ is continuous, so the sum does not apply.

$$\mathbb{E}[Z] = 0$$

Expectation and the long-run average

```
set.seed(6019)
rolls <- sample(1:6, 1e6, replace = TRUE) # a million die rolls
mean(rolls)
```

```
[1] 3.496123
```

The average of one million rolls is close to 3.5.

Properties of expectation

LINEARITY

For constants a, b, c and random variables X, Y :

1. $\mathbb{E}[c] = c$;
2. $\mathbb{E}[cX] = c \mathbb{E}[X]$;
3. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

These three properties are together called **linearity**. The expected value of the sample mean is derived from them alone.

Addition requires no independence

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

This holds for every pair X, Y .

Turnout and education are strongly dependent. The expected value of their sum is still the sum of their expected values.

Note: Variance behaves differently.

Expected utility

A decision model assigns a utility to each available action and predicts the action with the larger expected utility.

Jaywalking arrives sooner, worth 10. It carries a fine of 100 with probability .01. *Waiting* arrives later. It carries no fine.

$$\mathbb{E}[U_{\text{jaywalk}}] = 10 + .01(-100) = 9 \quad \mathbb{E}[U_{\text{wait}}] = 0$$

$\mathbb{E}[U_{\text{jaywalk}}] > \mathbb{E}[U_{\text{wait}}]$, so the model predicts *jaywalking*.

Expected utility in parameters

Let v be the value of arriving sooner; p_j and k_j the probability and cost of a fine while *jaywalking*; p_n and k_n the probability and cost of being stopped while *waiting*.

$$\mathbb{E}[U_{\text{jaywalk}}] = v - p_j k_j \quad \mathbb{E}[U_{\text{wait}}] = -p_n k_n$$

The model predicts *jaywalking* when $v - p_j k_j > -p_n k_n$.

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Center and spread

$\mathbb{E}[X]$ is the *balance point* of a distribution: $\mathbb{E}[X - \mathbb{E}[X]] = 0$.

Two distributions can share a value of $\mathbb{E}[X]$ and differ in how far X falls from that value. Variance is the expected value of $(X - \mathbb{E}[X])^2$.

Variance

VARIANCE

The **variance** of X is

$$\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_{i=1}^k p_i (x_i - \mu)^2$$

for a discrete X taking the values $x_1, \dots, x_k$ with $p_i = \mathbb{P}(X = x_i)$, and mean $\mu = \mathbb{E}[X]$. The **standard deviation** is $\sigma = \text{SD}[X] = \sqrt{\mathbb{V}[X]}$, so $\sigma^2 = \mathbb{V}[X]$.

Squared deviations from the mean, weighted by probability. The standard deviation returns the answer to the units of X .

Variance of a Bernoulli variable

$X \sim \text{Bernoulli}(p)$, with $\mathbb{E}[X] = p$:

$$\begin{aligned}\mathbb{V}[X] &= \sum_{i=1}^2 p_i (x_i - \mu)^2 \\ &= \underbrace{(1-p)}_{p_1} (\underbrace{0}_{x_1} - \underbrace{p}_{\mu})^2 + \underbrace{p}_{p_2} (\underbrace{1}_{x_2} - \underbrace{p}_{\mu})^2 \\ &= (1-p)p^2 + p(1-p)^2 \\ &= p(1-p) [p + (1-p)] \\ &= p(1-p)\end{aligned}$$

At $p = .6$, $\mathbb{V}[X] = .6 \times .4 = .24$ and $\text{SD}[X] = .490$.

Variance of a fair die

$$\mathbb{V}[X] = \frac{1}{6} \left[(1 - 3.5)^2 + (2 - 3.5)^2 + \dots + (6 - 3.5)^2 \right]$$

$$\mathbb{V}[X] = \frac{35}{12} \approx 2.917 \quad \text{SD}[X] \approx 1.708$$

The six squared deviations are 6.25, 2.25, 0.25, 0.25, 2.25 and 6.25. Their mean is $17.5/6 = 35/12$.

Properties of variance

PROPERTIES

For a constant c :

1. $\mathbb{V}[c] = 0$;
2. $\mathbb{V}[X + c] = \mathbb{V}[X]$;
3. $\mathbb{V}[cX] = c^2 \mathbb{V}[X]$.

Adding a constant moves the center of a distribution and leaves its spread unchanged.
Multiplying by a constant changes the spread by the square of that constant.

The computational identity

$$\begin{aligned}\mathbb{V}[X] &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] && \text{expand the square} \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 && \text{linearity; } \mu \text{ is a constant} \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 && \mathbb{E}[X] = \mu \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

“Variance equals the mean of the square minus the square of the mean.”

The computational identity on a fair die

$$\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\mathbb{E}[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$

$$(\mathbb{E}[X])^2 = 3.5^2 = \frac{49}{4}$$

$$\mathbb{V}[X] = \frac{91}{6} - \frac{49}{4} = \frac{182}{12} - \frac{147}{12} = \frac{35}{12} \approx 2.917$$

From Bernoulli to binomial

Distribution	$\mathbb{E}[X]$	$\mathbb{V}[X]$
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$

A binomial variable is a sum of n independent Bernoulli variables. Both columns of the second row are n times the first row.

Covariance

COVARIANCE

The **covariance** of X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] .$$

Properties of covariance

PROPERTIES

For constants a, b, c and d :

1. $\text{Cov}(X, X) = \mathbb{V}[X]$;
2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
3. $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$;
4. $\text{Cov}(X + c, Y + d) = \text{Cov}(X, Y)$.

The variance of a sum

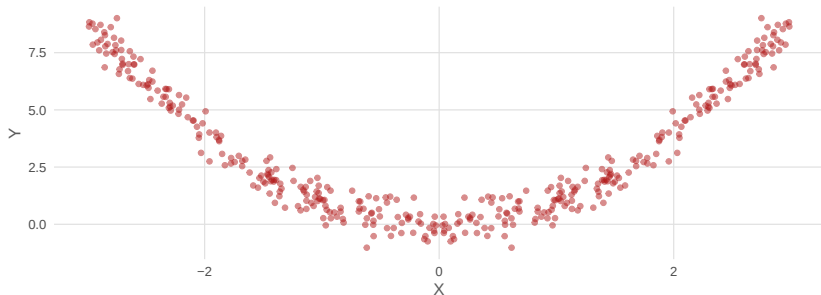
$$\mathbb{V}[X + Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2 \operatorname{Cov}(X, Y)$$

$$\mathbb{V}[X - Y] = \mathbb{V}[X] + \mathbb{V}[Y] - 2 \operatorname{Cov}(X, Y)$$

When X and Y are independent:

$$X \perp\!\!\!\perp Y \implies \operatorname{Cov}(X, Y) = 0 \implies \mathbb{V}[X \pm Y] = \mathbb{V}[X] + \mathbb{V}[Y]$$

Zero covariance is not independence



Here Y depends on X , but X is symmetric about zero, so $\text{Cov}(X, Y) = 0$.

$\text{Cov}(X, Y) = 0$ does not imply $X \perp\!\!\!\perp Y$. The implication runs in one direction only.

Roadmap

Discrete distributions

Continuous distributions

Data-generating processes

Conditioning

Expectation

Variance and covariance

Wrapping up

The operators

	Definition	Algebra
$\mathbb{E}[X]$	$\sum_i p_i x_i$	$\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, always
$\mathbb{V}[X]$	$\mathbb{E}[(X - \mu)^2]$	$\mathbb{V}[c] = 0$; $\mathbb{V}[X + c] = \mathbb{V}[X]$ $\mathbb{V}[cX] = c^2\mathbb{V}[X]$ $\mathbb{V}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
$\text{Cov}(X, Y)$	$\mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$	$\text{Cov}(X, X) = \mathbb{V}[X]$ $\mathbb{V}[X + Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\text{Cov}(X, Y)$

Further reading

- Gailmard, *Statistical Modeling and Inference for Social Science*, sections 2.2–2.3 on the conditional sample mean.
- Aronow and Miller, *Foundations of Agnostic Statistics*, ch. 1 on the expectation and variance operators.
- Blitzstein and Hwang, *Introduction to Probability*, ch. 4 (expectation and variance): <https://probabilitybook.net/>.